



Assessment of LRRK2 activity readouts in PPMI N370S GBA iPSCs and human neurons

Matthew J. LaVoie, University of Florida
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Summary

Variants in the glucocerebrosidase (GBA) gene are the most common genetic risk factor for Parkinson disease and Lewy Body Dementia. We, and others, have reported an interplay between GBA mutations and LRRK2 activity in cells. Here, we investigated LRRK2 activity readouts in patient-derived N370S GBA iPSCs and human neurons by protein biochemistry and immunoblotting.

Method

iPSCs were maintained in StemFlex™ medium (+StemFlex Supplement) until collection for protein analysis. Cells were collected in cold PBS and pelleted at 500g for 10 mins at 4C, and pellets frozen at -80C. Pellets were lysed in Cell Signaling™ Lysis buffer (cat no: 9803) [or 1% NP-40] supplemented with protease and phosphatase inhibitors for 30 mins on ice, centrifuged 21kg for 10 mins at 4C and cleared supernatants were analyzed for total protein by BCA. 20ug of each sample were analyzed by SDS-PAGE using standard immunoblotting protocols and commercial antibodies (pT72 Rab8a: ab230260; Rab8a: ab307607; pT73 Rab10: ab230261; Rab10: ab104859; pS106 Rab12: ab256487; Rab12: 18843-1-AP; GBA1: ab55080). Differentiation of iPSCs into cortical neurons was done using established protocols that are routinely used in the lab (Zhang et al, 2013; Sanyal et al., 2020). iPSCs were maintained in StemFlex media and transduced to express pTet-O-NGN2-puro and Fudelta GW-rtTA plasmids. Neuronal differentiation was achieved by culturing in selection media containing puromycin and doxycycline, inducing expression of NGN2 yielding ~99% purity. Detergent soluble fractions were solubilized in either 1% NP-40 or 1% Triton X-100. Insoluble fractions from iNs were prepared by solubilizing pellets in 2% SDS containing Cell Lysis buffer by sonication (3 pulses, 5 sec each, 30% intensity).

References

Zhang, Y. et al. Rapid Single-Step Induction of Functional Neurons from Human Pluripotent Stem Cells. *Neuron* 2013; 78; p785–798.

Sanyal A, Novis HS, Gasser E, Lin S, LaVoie MJ. LRRK2 kinase inhibition rescues deficits in lysosome function due to heterozygous GBA1 expression in human iPSC-derived neurons. *Frontiers in Neuroscience*, 2020;14:442, p.1-18.





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About the Authors

This document was prepared by Matthew J. LaVoie, Dept. of Neurology, CTRND, University of Florida. For more information, please contact Matthew J. LaVoie at (352) 273-5579 or by email at Matthew.LaVoie@neurology.ufl.edu

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